

RIWA KARAM

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Irvine, CA

Professional Summary

PhD candidate in Electrical and Computer Engineering specializing in multi-agent systems, robotics, machine learning, and optimization. Experienced in developing data-driven models, designing collaborative robotic systems, and translating research into practical solutions. Seeking research opportunities that advance intelligent autonomous systems through rigorous analysis, experimentation, and effective teamwork.

Industry Experience

- **ASML**  June 2025 – September 2025
Wilton, CT
Mechatronics Research Engineer Intern
 - Investigated machine health monitoring using statistical and Bayesian hierarchical models for uncertainty quantification and structural analysis.
 - Developed a Bayesian hierarchical model using Python and PyMC to predict drift and performance degradation, enabling root-cause analysis across machine lifetimes.
 - Built and compared decision tree, random forest, multiple regression, and baseline regression models to evaluate performance and identify patterns in the data.
 - Conducted data collection, preprocessing, and analysis using Python and MATLAB, including regression modeling, principal component analysis (PCA), and heatmap-based pattern recognition.
 - Applied Markov chain Monte Carlo (MCMC), Bayesian inference, and time-lag regression methods to characterize model uncertainty and temporal behavior.
- **ASML**  June 2024 – September 2024
Wilton, CT
Mechatronics and Control Systems Engineer Intern
 - Developed a Gaussian process regression (GPR) model using Python and MATLAB to predict force generated by a piezoelectric actuator with accuracy within ± 2 N; the work contributed to an ASML invention.
 - Built regression and machine-learning models to identify correlations and dependencies in experimental data.
 - Automated and merged MATLAB and Python scripts that run testing workflows for both the mechatronics and software teams, reducing runtime from 40 hours to an overnight run.
 - Investigated feasibility of optimization problems and current-control hardware issues for test stands by analyzing their electrical circuitry and alternative hardware.
- **AviaPro Consulting Inc.**  June 2022 – July 2023
Remote
Software Developer
 - Developed web applications using React.js and Node.js, replacing Excel-based repositories with streamlined interfaces connected to multiple databases.
 - Applied software-development and DevOps practices, including CI/CD, infrastructure as code, and Git.
 - Contributed to team-based full-stack projects and independently developed an internal full-stack application.
 - Transitioned to a part-time role in September 2022 after completing a summer co-op.

Academic Experience

- **UCI Robot Ecology Laboratory**  September 2023 - Present
Irvine, CA
Graduate Student Researcher
 - Conducting research in collaborative multi-agent systems, optimization, machine learning, ecology-inspired robotics, and human-swarm interaction under the supervision of Professor Magnus Egerstedt and Professor Yanning Shen.
 - Maintaining the Robot Ecology Lab Robotarium testbed through robot debugging, repair, and system troubleshooting.
 - Presented a 1st-author regular paper at the 64th IEEE Conference on Decision and Control (CDC) in Brazil.
 - Presented ongoing research at ARM, Inc. in Austin, TX under the Robotalk External Talk Series, invited by Dr. Federico Pecora.
 - Presented ongoing research at the 45th Southern California Control Workshop in UCSD.
 - Passed my PhD qualifying examination in May 2026 and advanced to PhD candidacy.
 - Won the Software & Algorithms track at the 2026 UCI GEECS Annual Technology Showcase.
 - Presented a poster on my master's thesis at the 2025 UCI GEECS Annual Technology Showcase.
 - Mentored undergraduate researchers and collaborated with them on research projects.
 - Led laboratory Robotarium testbed tours and live multi-robot demonstrations for more than 100 visitors.
 - Co-authored the UCI Robot Ecology Lab manual, documenting debugging procedures, experiment workflows, and Vicon system maintenance.

Education

- **University of California, Irvine** [🌐] September 2023 - Expected June 2027
Electrical & Computer Engineering, Doctor of Philosophy Irvine, CA
 - Dissertation: Collaboration in Multi-Agent Systems: From Assignment to Learning
 - Advisors: Professor Magnus Egerstedt, Professor Yanning Shen
- **University of California, Irvine** [🌐] September 2023 - June 2025
Electrical & Computer Engineering, Master of Science Irvine, CA
 - GPA: 3.975/4.00
 - Thesis: A Graphical Interface for Specifying and Establishing Multi-Robot Formations
 - Relevant Coursework: Optimization, Network Science, Control and Machine Learning, Linear Systems
- **University of Balamand, Al-Kurah** [🌐] August 2020 - June 2023
Computer Engineering, Bachelor of Science Balamand, Al-Kurah, Lebanon
 - GPA: 3.90/4.00
 - Computer Engineering Class of 2023 Valedictorian
 - Graduation Project: Gamification of Virtual Reality Kitchen Scenarios
 - Tutoring: Python, MATLAB, C, signals and systems, microcontrollers, embedded systems, computer architecture.

Publications

C=CONFERENCE, T=THESIS, R=UNDER REVIEW, S=SUBMITTED

- [S.1] Riwa Karam, Ian Kuo, and Magnus Egerstedt, **Freehand Sketching for End-User Programming of Robot Swarms**. arXiv preprint [DOI](#)
- [R.1] Riwa Karam, Ruoyu Lin, Brooks A. Butler, and Magnus Egerstedt, **Learning Altruistic Collaboration in Heterogeneous Multi-Team Systems**. Under Review in the IEEE Transactions on Robotics (T-RO). arXiv preprint [DOI](#).
- [R.2] Riwa Karam, Alexander A. Nguyen, Ruoyu Lin, David R. Martin, Diana Morales, Brooks A. Butler, and Magnus Egerstedt, **Collaboration in Multi-Robot Systems: Taxonomy and Survey over Frameworks for Collaboration**. Under Review in the IEEE Control Systems Magazine (CSM). arXiv preprint [DOI](#).
- [C.1] Riwa Karam, Ruoyu Lin, Brooks A. Butler, and Magnus Egerstedt, **Resource Allocation for Multi-Team Collaboration Based on Hamilton's Rule**, IEEE 64th Conference on Decision and Control (CDC), Rio de Janeiro, Brazil, 2025, pp. 6891-6898, [DOI](#).
- [T.1] Riwa Karam, **A Graphical Interface for Specifying and Establishing Multi-Robot Formations**, M.S. Thesis, University of California, Irvine, 2025, [Official Record](#).

Skills

- **Programming Languages and Libraries:** Python, MATLAB, scikit-learn, PyTorch, Keras, NumPy, pandas, Matplotlib, shell scripting, PyMC, C, Java, C#.
- **Hardware Technologies:** Robot testbeds, Vicon Motion Capture (MoCap) System, robot building & debugging.
- **Other Tools & Technologies:** Git & GitHub, LaTeX, VS Code, Unity, Microsoft Office, Google Workspace.

Selected Honors and Awards

- **GEECS Annual Technology Showcase First Place Winner** June 2026
Software & Algorithms Track, University of California, Irvine [🌐]
- **Faculty Award for Academic Excellence** July 2023
University of Balamand, Faculty of Engineering [🌐]
- **First Place Winner, Lebanon IoT & AI Challenge 2021** November 2021
Arab IoT & AI Challenge [🌐]
- **Dean's Honor List** August 2020 - June 2023
University of Balamand, Faculty of Engineering

Professional Service

- **UCI Society of Graduate Electrical Engineers and Computer Scientists** [🌐] June 2026 – Present
Web Content Manager Irvine, CA
- **IEEE Control Systems Society NextCom** [🌐] March 2026 – Present
General Activities Committee Member Remote

Languages

- Arabic: Native proficiency
- English: Full professional proficiency
- French: Full professional proficiency
- Spanish: Elementary proficiency (CEFR A2)